QPUBench

Open Source Quantum Computing Benchmark Framework

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Molecular Quantum Solutions (MQS)

Framework Overview

What is QPUBench?

Modality-agnostic quantum benchmark framework — three layers, one record

Layer	Answers	Lives in
Schema	*What* are you benchmarking?	`schemas/` — 38 Pydantic v2 modules (= "Schema v3.0.0")
Adapter	*How* does it run?	`backends/` + `integrations/` (next slide)
Store	*How* is it persisted?	`store.py` — `NDJSONStore`, `ParquetStore`, `S3Store`

Core invariants

qpubench itself never imports from any quantum library — only adapters do. And since schema v3.0.0 the core schema modules import *zero vendor schema modules*: vendor data flows through vendor-neutral dict extension points (next section).

Architecture

Core layer *(no SDK deps, `pip install`-able)*

- `schemas/` — Schema layer
- `runner.py` — `BenchmarkRunner`
- `store.py` — Store layer

Integration layer *(reference code)*

- `backends/` — adapter protocols + stubs
- `integrations/` — real reference adapters, **not installed** — copy into your project (like a template)
- `examples/` — runnable demos

"Contract" = a Python `Protocol` — a fixed method signature (`spec`/`validate`/`run`) callers rely on without caring which class implements it. A **stub** satisfies it with fake data; real adapters (Aer, IBM, IQM, Braket) satisfy the same contract with real SDK calls. Swapping one for the other needs zero code changes elsewhere.

Key Design Principles

Principle	Implementation
Schema-first	Pydantic v2 models are the stable contract — every record round-trips through `.model_dump_json()`
No SDK lock-in	Core never imports any quantum library
Three adapter protocols	`BackendAdapter` · `AlgorithmAdapter` · `ErrorMitigationAdapter`
Append-only stores	`NDJSONStore`/`ParquetStore` for reproducible sweeps; `S3Store` for concurrent writers
Language-agnostic	Every record is plain JSON — any language can parse it, whichever store wrote it
Inputs vs. outputs	`VQAConfig` = what you chose to run; `VQAResult` = what the run produced — computed values are never user-supplied
Vendor-neutral core	Vendor payloads live in keyed dicts (`vendor_results`, `mitigation_options`, `vendor_data`) — adding a vendor never touches a core file

NDJSON vs. JSON, Precisely

Every `BenchmarkRecord` serializes to one JSON object, always. *How that object reaches disk* differs by store — same schema, three storage strategies, not an inconsistency:

Store	On-disk shape
`NDJSONStore`	One JSON object per line, appended to one shared file ("**N**ewline-**D**elimited JSON")
`S3Store`	One whole JSON file per record — no shared file, so concurrent writers can't race
`ParquetStore`	Same records converted to a columnar table

Installation

Not yet on PyPI — install from a git clone until it is:

```
git clone https://github.com/mqsdk/qpubench && cd qpubench
pip install .                # Minimal — schemas + stubs, no quantum SDK
pip install ".[qiskit]"      # Qiskit Aer + IBM Quantum Runtime V2
pip install ".[storage]"     # Parquet store (pyarrow + pandas)
pip install ".[s3]"          # S3 store (boto3) — AWS S3 / MinIO / HF buckets
pip install ".[all]"         # everything, all optional extras
```

Package managers: `pip` · `uv` · `Poetry 2` · `conda`

Credentials: `.env` + `BackendSpec.auth`

Schema Layer

Reading This Reference

The next 9 slides are a **reference listing**, not a narrative — they exist so you know a module is there when you need it, not to be read top-to-bottom. The take-home message fits in one line:

Take-home message

Every schema module is a set of typed Pydantic models for one vendor/technique's inputs and outputs. 11 "core" modules are framework-owned (no vendor); the rest are one module per integration, named so the filename alone tells you who maintains it.

If you just want to see the package in action: skip ahead to *Adapters* (real running code) or *Examples & Coverage* at the end (a full worked example). Come back here when you need to look up what a specific module contains — that's what a GitHub repo's own reference docs (`docs/schemas.md`) are for; these slides just mirror the module list so it's not a total blank spot.

Core Schema Modules (1/3)

All rows are computing-model/qubit-modality **all / all** — these are the framework's own foundational types, not vendor-specific.

Module	Key types
`primitives`	`ComputingModel`, `QubitModality`, `CircuitFormat`, `PauliLabel`, `ComplexNumber`
`circuit`	`CircuitSpec` — `from_openqasm3()`, `bind()`, `is_parametric()`
`observable`	`SparsePauliObservable`, `PauliTerm`
`backend`	`BackendSpec` — 28 factory constructors

Core Schema Modules (2/3)

Module	Key types
`execution`	`ExecutionOptions`, `AlgorithmSpec`, `ZNEConfig`
`advantage`	`QuantumAdvantageRecord` — multi-org registry, unprefixed
`result`	`QuantumResult` (+ `vendor_results` dict), `ExpectationResult`, `ShotResult`
`record`	`BenchmarkRecord`, `VQAConfig` (inputs), `VQAResult` (computed outputs)

Core Schema Modules (3/3)

Module	Key types
`optimizer_catalog`	`MINIMIZER_CATALOG`, `STOPPING_CRITERION_CATALOG` — over `AdaptVQEConfig` fields
`hamiltonian_library`	`HamiltonianSource`, `HamiltonianLibraryRecord` — metadata for real Hamiltonians (PennyLane qchem, HamLib, ab initio)
`contraction_path`	`ContractionPathStrategy/Config/Result` — real quimb + cotengra

Domain-Specific Schema Modules (1/2)

Named `<org>_<package>.py` — who maintains the upstream project + what it's called.

Module	Computing model (qubit modality)
`johnrscott_mbqc_fpga`	MBQC-FPGA — bit-exact 16-bit program word
`dtu_photonic`	LOQC / FBQC (photonic) — Fock states, HOM, photonic VQE/analog
`dtu_gbs`	GBS (photonic) — hafnian, vibronic, TDM/Borealis
`microsoft_qdk`	Gate-based (QPE technique) — SCF $\rightarrow$ active space $\rightarrow$ resource estimation
`mqsdk_qse`	Gate-based (KQD technique) — Krylov Quantum Diagonalization / SQD

Domain-Specific Schema Modules (2/2)

`classiq_classiq`/`pyscf_pyscf` double up their name because the maintaining org and the package name are the same word — kept for consistency rather than shortened. `polarizable_embedding` stays unprefixed: it bridges CPPE + PyFraME with no single vendor to name.

Module	Computing model (qubit modality)
`molssi_qcschema`	all / all — QCSchema / QCElemental / PennyLane interoperability
`quera_bloqade`	Adiabatic (neutral atom) — Bloqade / Aquila AHS
`erikkjellgren_slowquant`	Gate-based — SlowQuant UCC/VQE ansatz, SCF, linear response
`classiq_classiq`	Gate-based — Classiq synthesis + chemistry/QAOA
`pyscf_pyscf`	Gate-based (chemistry) — mean-field/DFT, PCM solvation, ERI builder
`polarizable_embedding`	Gate-based (chemistry) — CPPE + PyFraME, via `pyscf.solvent.PE`

Computing Model × Qubit Modality

Two **independent, orthogonal** axes — not one flat enum (a fixed set of named choices). Many-to-many: some paradigms run on several QPU modalities, but not every pairing exists.

`ComputingModel` (paradigm)

- `GATE_BASED` — circuit model
- `MBQC` — measurement-based
- `FUSION_BASED` — FBQC resource states + fusion
- `ADIABATIC` — continuous-time evolution (AHS)
- `ANNEALING` — quantum annealing
- `GBS` — Gaussian Boson Sampling
- `SAMPLING` — general sampling paradigms

`QubitModality` (QPU modality)

- `SUPERCONDUCTING` — IBM, IQM
- `TRAPPED_ION` — Quantinuum, IonQ
- `NEUTRAL_ATOM` — QuEra
- `PHOTONIC` — Quandela, Xanadu
- `SILICON_SPIN` — Quantum Motion

`GATE_BASED` alone spans superconducting, trapped-ion, silicon-spin, and photonic hardware — `GBS`/`FUSION_BASED` are photonic-only today. `QPE`/`KQD` are techniques on `GATE_BASED`, not separate paradigms.

CircuitSpec — One Type for All Circuits

```
# Gate-based — read a .qasm file straight into `serialized`
source = pathlib.Path("bell_state.qasm").read_text()
circuit = CircuitSpec(num_qubits=2, serialized=source,   # OpenQASM 2.0
                      observables=[...])

# OpenQASM 3.0 — same file-read pattern
circuit = CircuitSpec.from_openqasm3(
    pathlib.Path("bell_state.qasm3").read_text(), num_qubits=2)

# Parametric VQE — bind() returns a new copy
ansatz = CircuitSpec(num_qubits=2, parameters=["theta"], ...)
bound  = ansatz.bind({"theta": 1.2566})
```

Schema v3.0.0 — Inputs vs Outputs, Vendor-Neutral Core

Two breaking changes define v3.0.0. **First**: VQA metadata is split — configs never carry computed values (passing one raises a `ValidationError`):

```
vqa = VQAConfig(problem_type="chemistry", molecule="H2", basis="sto-3g")  # inputs
record = runner.run(bound_ansatz, "aer", shots=4096, vqa=vqa)
record.vqa_result.final_eigenvalue    # derived from expectation values by the runner
record.vqa_result.chemical_accuracy   # True (< 1.6 mHa) once a reference is present
```

Second: core schemas import zero vendor modules — vendor payloads travel as keyed dicts (pass a Pydantic model, it's auto-dumped; rehydrate with the vendor schema):

Extension point	Replaces	Example key
`QuantumResult.vendor_results`	24 vendor-typed result fields	`"qforte_result"`, `"slowquant_record"`
`CircuitSpec.measurement_pattern` / `.photonic_circuit`	typed MBQC/photonic fields	`MBQCPattern.model_validate(...)`
`ExecutionOptions.mitigation_options`	QESEM-typed fields	`qesem_mitigation_options(...)`
`VQAConfig.vendor_data`	ad-hoc vendor fields	`{"cebule": {...}}`

Real Hamiltonian Sources

Three ways to get a real molecule's qubit Hamiltonian as a `SparsePauliObservable` (`hamiltonian_sources.*`) — drops into any Estimator path:

Source	Function	What it gives you
HamLib Chemistry	`load_hamlib_chemistry()`	HDF5 (NERSC), 4 encodings, no ref. energies
PennyLane qchem	`load_pennylane_qchem()`	Dataset service, ships `fci_energy/vqe_energy`
Ab initio	`build_qubit_hamiltonian()`	*Any* geometry — PySCF HF + active-space + mapper

```
obs, record = build_qubit_hamiltonian(
    geometry, basis="sto-3g", active_electrons=2, active_orbitals=2)
```

Verified: HamLib's H2 Hamiltonian through the *unmodified* ADAPT-VQE engine converges to `-1.13146 Ha`, matching dense diagonalization to `7e-15`.

Tensor-Network & Chemistry Mechanisms

Real, verified mechanisms built on `quimb+cotengra` or PySCF (already a dependency):

Mechanism	Implementation	Verified
Contraction Path Finder	`quimb.tensor.Circuit` + `cotengra.HyperOptimizer`	4/4 strategies real; `NONE` costs more (`180224` vs `248`)
ERI Builder	`mol.intor('int2e')` vs `mf.density_fit()` (RI/DF)	RI matches standard ERIs to `2e-5` Ha (H2O/cc-pVDZ)
Gate Selector	`GateSelector` protocol — gradient screen vs full re-opt	Both converge to exact H2 ground state
Orbital Optimizer	Newton (`CASSCF`) / Simple / Basin-Hopping	All 3 agree on H2/6-31G to `3e-9` Ha

A real bug found along the way: quimb's `contraction_info(optimize=False)` raises `TypeError` — worked around via `opt_einsum.contract_path()` directly.

`reactions` — Bridging Cantera, PennyLane & Cebule

Moved out of "core" — it bridges three real external tools, not a shape qpubench invented:

Bridges to	What it adds
Cantera	`ReactionMechanism.to_cantera_yaml()` — real, loadable mechanism
PennyLane demo	`ReactionPathResult.rate_constant()` — barrier → classical `k = A exp(-Ea/RT)`
Cebule `RXN_OPT`	Complementary: network flux optimization, not merged in

Verified against real `cantera==3.2.0`, all 3 reaction types. Real gotcha found: Cantera's default `quantity` unit is `kmol` not `mol` — an unstated `Ea` silently becomes 1000x too small; `to_cantera_dict()` always declares units explicitly to prevent it.

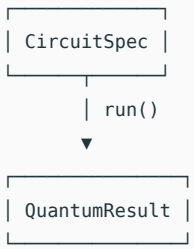
Adapters

Adapters

What vs. how: this slide is the *what* — three abstract shapes, no vendor code yet. Every slide after is the *how* — a vendor satisfying one shape with real code.

Three Adapter Protocols

`BackendAdapter` *Circuit in, result out*

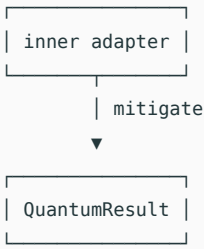


Aer · PennyLane · IBM · IQM · Braket · MBQC

`AlgorithmAdapter` *Problem spec in*

QForte · SlowQuant · generic ADAPT-VQE

`ErrorMitigationAdapter` *Wraps a BackendAdapter*



Mitiq (real, next section) Fire Opal/Haiqu (schema only)

`BenchmarkRunner` dispatches automatically — `ErrorMitigationAdapter` registers like any `BackendAdapter`.

BackendAdapter — Estimator vs Sampler

```python
class MyBackendAdapter:
    @property
    def spec(self) -> BackendSpec: ...    # hardware description
    def validate(self, circuit) -> list[str]: ...
```

Estimator path — `circuit.observables` set

```python
def run(self, circuit, options):
    evs = [measure_expectation(circuit, obs)
           for obs in circuit.observables]
    return QuantumResult(
        expectation_values=evs, ...)
```

VQE / QAOA objective values

Sampler path — no observables

```python
def run(self, circuit, options):
    counts = sample_bitstrings(
        circuit, options.shots)
    return QuantumResult(
        shots=ShotResult(
            ..., counts=counts), ...)
```

Raw bitstring / shot counts

Rule: SDK imports go inside the method that uses them — never at module level.

AlgorithmAdapter (QForte example)

```
class QForteAlgorithmAdapter:
```

validate_problem()

```
def validate_problem(self, circuit):
    if circuit.format != \
            CircuitFormat.MOLECULE_JSON:
        return [
            f"Expected MOLECULE_JSON, "
            f"got {circuit.format!r}"
        ]
    return []
```

run_algorithm()

```
def run_algorithm(self, circuit, options):
    import qforte as qf    # deferred import
    cfg = options.adapt_vqe_config
    mol = qf.system_factory(...)
    alg = qf.ADAPTVQE(mol, ...)
    alg.run(pool_type=cfg.pool_type,
            optimizer=cfg.optimizer)
    result = QuantumResult(
        vendor_results={"qforte_result":
            QForteRunResult(...)},
        expectation_values=[...])
    return (result, VQAConfig(...),
        VQAResult(final_eigenvalue=
            alg.get_gs_energy(), ...))
```

AlgorithmFamily — Switch Implementations Freely

`AlgorithmSpec` carries **identity only** (`name` + `family`) — hyperparameters live in one shared config every adapter for that family accepts:

```
config = AdaptVQEConfig(pool_type="SD", optimizer="BFGS")
opts = ExecutionOptions(algorithm_spec=AlgorithmSpec(
    name="ADAPTVQE", family=AlgorithmFamily.ADAPT_VQE),
    adapt_vqe_config=config)

runner.run(mol, "qforte", opts)                  # native C++
runner.run(mol, "ibm_qiskit_adapt_vqe", opts)     # from-scratch Qiskit
runner.run(mol, "microsoft_qdk_adapt_vqe", opts)  # QDK / Azure Quantum
```

Three adapters implement `AlgorithmFamily.ADAPT_VQE`: `integrations/qforte/` (native C++), and `ibm_qiskit_adapt_vqe/`/`microsoft_qdk_adapt_vqe/` (thin wrappers over the pure-Python `generic_adapt_vqe/` engine).

Circuit Optimization — Three Real Strategies, Not Two

Every gate-based backend adapter already runs the **Qiskit transpiler** (`ExecutionOptions.optimization_level` 0–3 + `TranspilerConfig`) — the baseline every circuit gets. Xenakis/Classiq below are two *additional*, heavier strategies for when that isn't enough.

	Qiskit transpiler	Xenakis	Classiq
Strategy	Preset pass manager	GA **search**	Constrained **synthesis**
Objective	`optimization_level` 0–3	Soft penalty ($\lambda_{\text{depth}}, \lambda_{2q}$)	Hard bound + objective
Cost	One pass, ms	Generations × population	One solve (duration, s)

Not yet modeled: Microsoft QDK/Q#'s own circuit-optimization passes have no schema representation here — `microsoft_qdk.py` stops at resource estimation. Open gap, not silently assumed.

Xenakis ↔ Classiq — Comparing Two Optimization Runs

```
classiq_mol = ClassiqMoleculeSpec.from_xenakis_molecule(xenakis_mol)
cmp = CircuitOptimizationComparison(ga_result=ga_run, classiq_result=synth)
cmp.depth_delta          # GA best-genome depth − Classiq depth
cmp.search_cost_label    # "GA: 40 generations vs Classiq: 2.30s synthesis"
```

Error Mitigation Providers

One module per vendor, one shared `ErrorMitigationStrategy` enum in `primitives.py`:

Provider	Schema module	Technique
Q-CTRL Fire Opal	`qctrl_fire_opal`	Closed-box noise suppression
Mitiq	`unitaryfund_mitiq`	ZNE · PEC · CDR · REM · DDD
Haiqu Rivet	`haiqu_rivet`	HW-aware transpilation caching
ParityQC	`parityqc_parityqc`	Parity encoding (QUBO/HCBO/Ising)
QMatter	`qmatter_qmatter`	Quantum problem compression
Quantum Motion	`quantum_motion_hardware`	CMOS spin-qubit characterisation *(hardware vendor)*
QESEM (Qedma)	`qedma_qesem`	QET noise scaling + characterisation

ErrorMitigationAdapter — Demonstrated (Mitiq ZNE)

Only **Mitiq** has a real `ErrorMitigationAdapter` behind it today (`MitiqZNEAdapter`) — the rest on the previous slide are schema-only metadata. Wraps any `BackendAdapter`; mitigates the estimator path only.

```
noisy_inner = AerAdapter(noise_model_json=depolarizing_2pct_5pct_json)
raw = noisy_inner.run(bell_circuit, ExecutionOptions())
raw.expectation_values[0].value             # 0.950  (noisy)

adapter = MitiqZNEAdapter(noisy_inner)      # same BackendAdapter contract
mitigated = adapter.run(bell_circuit, ExecutionOptions())
mitigated.expectation_values[0].value       # 0.997  (extrapolated)
```

Verified against a real depolarizing-noise Aer simulator (2%/5% error on H/CX): ZNE recovers `0.997` vs. raw noisy `0.950` and noiseless-exact `1.0` — real Mitiq `Factory.run()/.reduce()` calls, the adapter `base.py` described but that never existed until now.

Running on Real Hardware — IBM, IQM & AWS Braket

All three adapters are **real, verified end-to-end** — Aer/Braket need no credentials (`BraketLocalBackend`); IBM/IQM verified against bundled fake backends. `opts = ExecutionOptions(shots=4096)` shared by all three.

IBM Quantum Runtime V2

```
adapter = IBMAdapter(
    "ibm_torino",
    execution_mode=
        IBMExecutionMode.SESSION)
runner.register(adapter, "ibm")
runner.run(circuit, "ibm", opts)
```

Estimator/Sampler auto-chosen.

IQM Resonance cloud

```
adapter = IQMAdapter("garnet",
    token=os.environ["IQM_TOKEN"])
runner.register(adapter, "iqm")
runner.run(circuit, "iqm", opts)
```

Star topology, no SWAP routing.

AWS Braket

```
spec = BackendSpec.braket(
    "arn:...Ankaa-3",
    s3_bucket_ref="BUCKET")
adapter = BraketAdapter(
    spec.auth["device_arn"],
    s3_bucket_ref="BUCKET")
runner.register(adapter, "braket")
```

Needs an S3 result location.

All three implement `TranspilableBackend`. Two real SDK migrations found while verifying: `qiskit-iqm` standalone is obsolete (`iqm-client[qiskit]` replaces it); IBM's old `channel="ibm_quantum"` is gone (`"ibm_quantum_platform"` replaces it).

IBM Quantum Advantage Tracker

Problem: "quantum advantage" claims are hard to compare across papers — no shared record of what ran, on what hardware, vs. what classical method. `BenchmarkRecord.advantage` holds one `QuantumAdvantageRecord`, attached after the fact:

```
record = runner.run(circuit, "ibm_torino", options)
record = record.model_copy(update={"advantage": QuantumAdvantageRecord(
    experiment_type=AdvantageExperimentType.OBSERVABLE_ESTIMATION,
    circuit_name="Loschmidt-echo-70q", num_qubits=70, backend_name="ibm_boston",
    observable_value=0.327, observable_error_bound=0.012,
    classical_method=ClassicalComparisonMethod.TENSOR_NETWORK,
    submission_url="https://github.com/quantum-advantage-tracker/...",
)})
```

Quantum Advantage Tracker: community registry co-initiated by IBM, Flatiron Institute, BlueQubit, Algorithmiq. Three experiment categories: observable estimation · variational · classically verifiable.

Execution & Storage

BenchmarkRunner

```
runner = BenchmarkRunner(store=NDJSONStore(Path("results.ndjson")))
runner.register(AerAdapter(), name="aer")
runner.register(QForteAlgorithmAdapter(), name="qforte")
record = runner.run(circuit, "aer", ExecutionOptions(shots=4096))   # single run

# Cartesian product sweep: circuits x backends x options
records = runner.sweep(
    circuits=[circuit_h2, circuit_lih], backend_names=["aer", "qforte"],
    options_list=[ExecutionOptions(shots=1024), ExecutionOptions(shots=8192)],
    run_id="vqe-sweep-001",
)
runner.add_hook(lambda rec: log_energy(rec.vqa_result.final_eigenvalue))   # post-run hook
```

Structured Logging — `BenchmarkLogger`

Built directly on `BenchmarkRunner.add_hook()` — real levels, handlers, and formatters, not one flat log line:

```
from qpubench import BenchmarkLogger
bench_logger = BenchmarkLogger(name="qpubench.benchmark")
bench_logger.attach(runner)          # wires into add_hook() internally
record = runner.run(circuit, "aer", ExecutionOptions(shots=4096))
# {"experiment_id": "...", "backend": "aer_statevector", "status": "succeeded", ...}
```

Levels	Handlers	Formatters
SUCCEEDED → `INFO`, FAILED → `ERROR`	any `logging.Handler`	`JSONFormatter` (default), swappable

Data Persistence

NDJSONStore — zero dependencies, append-only, grep-able

```
store = NDJSONStore(Path("results/run1.ndjson"))
store.save(record)
record  = store.load(experiment_id)           # by UUID
records = store.query(backend__name="aer_statevector", result__status="succeeded")
```

ParquetStore — columnar analytics (`pip install ".[storage]"`)

```
store = ParquetStore(Path("results/run1.parquet"))
df = store.to_dataframe()                     # → pandas DataFrame
```

Every record is one JSON line — stream-able, pandas-ready, archive-safe.

S3Store — AWS S3 & Hugging Face Buckets

One JSON object per record (`{prefix}/{experiment_id}.json`) — no shared file, so concurrent writers never race. `pip install "qpubench[s3]"`.

AWS S3 (S3-compatible too)

```
store = S3Store(
    "my-results-bucket",
    prefix="sweeps/2026-07-06",
    region_name="eu-west-1",
)
```

Any `boto3.client("s3", ...)` kwarg works — same path covers MinIO too.

Hugging Face Buckets

```
store = S3Store.huggingface(
    bucket="my-training-bucket",
    namespace="my-username",
    access_key_id="HFAK...",
)
```

`.huggingface()` sets the gateway's required `Config`.

Same `ResultStore` protocol as `NDJSONStore`: `save()` / `load()` / `query()` / `all()`.

Integrations

Integration Ecosystem — `BackendAdapter`

Schema modules are named `<org>_<package>.py` — one glance tells you who maintains it and what it's called.

Integration	Schema module
Cebule SDK	`mqsdk_cebule`
Photochipsim / FBQC	`dtu_photonic`
DTU-GBS / photonic_QC	`dtu_gbs`
QESEM (Qedma)	`qedma_qesem`
Bloqade / Aquila	`quera_bloqade`
IQM Resonance (garnet · deneb · sirius)	—
Quantum Motion CMOS spin-qubit	`quantum_motion_hardware`

Integration Ecosystem — `AlgorithmAdapter`

Integration	Schema module
Xenakis GA circuit search	`mqsdk_xenakis`
QForte (ADAPT-VQE, UCCN-VQE, …)	`evangelistalab_qforte`
Classiq circuit synthesis	`classiq_classiq`
ExcitationSolve (Fourier VQE)	`dlr_excitation_solve`
GSOpt (ground-state benchmark)	`bestquark_gsopt`
QDK / QuNorth chemistry	`microsoft_qdk`
QSE / KQD	`mqsdk_qse`
SlowQuant UCC/VQE (`integrations/slowquant/`, real code, not on PyPI)	`erikkjellgren_slowquant`

Integration Ecosystem — `ErrorMitigationAdapter`

One module per vendor now — previously all bundled into a single `error_mitigation` grab-bag (including a hardware vendor miscategorized as "error mitigation").

Integration	Schema module
Q-CTRL Fire Opal	`qctrl_fire_opal`
Mitiq (ZNE · PEC · CDR · REM · DDD)	`unitaryfund_mitiq`
Haiqu Rivet transpiler	`haiqu_rivet`
ParityQC parity encoding	`parityqc_parityqc`
QMatter problem compression	`qmatter_qmatter`

What Is Kubeflow, and Why Is It Here?

For readers who haven't met Kubeflow: an open-source platform that runs a workflow as a graph of steps (a DAG) on Kubernetes — each step is a container; it schedules them, tracks inputs/outputs, and shows the graph in a web dashboard.

```
mol_map ⟶ tn_qc_opt ⟶ qasm_gen ⟶ execute_circuits
  (box)       (box)       (box)         (box)
```

Why it's here: earlier integrations only needed `runner.run()`. Kubeflow becomes useful once a *sequence* of qpubench steps needs to run unattended, on a schedule, with a visual record — e.g. a nightly benchmark sweep. The next slides show that sequence as a real, compiled Kubeflow pipeline.

Orchestration — Kubeflow Pipelines

Different from every integration above: not a schema bridge, but how algorithmic steps get **run** as scheduled, dashboard-visible components.

Two Kubeflow SDKs, two roles:

SDK	Role here
`kfp` (Pipelines)	Owns the DAG + Central Dashboard — every step is a native `@dsl.component`
Unified SDK	Narrow, opt-in (`TrainerClient`/`OptimizerClient`) — called *inside* one component body only, for multi-node dispatch or Katib search

Why not the unified SDK as the top-level abstraction?

Its own docs list Pipelines support as "Planned," not shipped, and `TrainerClient`/`CustomTrainer` are shaped around ML training. `kfp` already gives DAG construction, branching, and caching natively.

Component Taxonomy

Every pipeline decomposes into two kinds of `kfp` node — same shape, different resource profile:

Kind	Shape	Examples
Transform	pydantic-in → pydantic-out, no `BackendAdapter`	SCF, active-space selection, Cebule MOL_MAP/TN_QC_OPT
Execution	`BenchmarkRunner.run()` against a registered adapter	GPU sim (Qrack) · real QPU (IBM, IQM, QESEM)

Kubeflow can't reach a QPU directly — an Execution component does what `BackendAdapter.run()` already does: call the vendor SDK, submit to *their* queue, poll, return. Kubeflow's role is a scheduled, credentialed place to host that client, not compute.

Iterative loops stay one job — `n_iterations`/macro-micro-iterations never explode into one DAG node per iteration.

Worked Example — Cebule Task Chain as Four `kfp` DAGs

`integrations/kubeflow/` — one `@dsl.pipeline` per `Mapper` category in `data/IBM_VQE_Test_Benchmark.csv`, matching Cebule SDK's own documented task order:

```
tn_qc_opt+mol_map:  mol_map → tn_qc_opt → qasm_gen → execute_circuits
tn_qc_opt:          jw_map  → tn_qc_opt → qasm_gen → execute_circuits
mol_map:            mol_map ──────────────────────→ execute_static_hamiltonian
JW:                 jw_map  ──────────────────────→ execute_static_hamiltonian
```

`mol_map`/`JW` carry no VQE ansatz — they measure the fixed Hartree-Fock reference state instead.

Verified by actually compiling all four against `kfp==2.16.1`: surfaced a real gotcha — `from __future__ import annotations` breaks kfp's component introspection (needs live type objects, not PEP 563 strings). Every component body was also run directly (`.python_func`), including a real PySCF/OpenFermion `jw_map` call. Credentials injected via `kfp-kubernetes`'s `use_secret_as_env`, never as plain pipeline parameters.

Kubeflow — Sweep Design

Rule: `Mapper` change → new pipeline (different components). Every other benchmark column is a *parameter value*, resolved inside one pipeline run — never one DAG per CSV row:

Column	Resolved via
`TN_Layers_Network/Circuit`, `Rotation_Type`, `Measurement_Method`	nested `dsl.ParallelFor` — one dashboard graph, fanned out
`Shots`, `Qiskit_Opt_Level`	`BenchmarkRunner.sweep()` *inside* the Execution component — same "stays one job" treatment as `n_iterations`

Conversion utilities — closed TODO

`SparsePauliObservable.to_dense_matrix()` / `.from_dense_matrix()` now ship in core (verified against Qiskit's `SparsePauliOp`, size-guarded via `max_qubits`) — all 6 (hamiltonian_kind, measurement_method) branch combinations run; only the live Cebule `create_task` calls still await real credentials.

Cross-Framework Compatibility

Pauli encoding — explicit converters prevent silent convention bugs:

```
# Qrack Q# convention: I=0, X=1, Z=2, Y=3  (non-sequential!)
PauliLabel.Z.to_qrack_int()           # → 2  (not 3)
PauliLabel.Y.to_qrack_int()           # → 3  (not 2)

# Qiskit C API QkBitTerm bit-packed encoding
PauliLabel.X.to_qiskit_c_bit_term()   # → 0b0010

# MBQC byproduct register: bit 0 = Z, bit 1 = X  (reversed vs gate-based!)
```

Complex numbers: stored as `{"re": 1.0, "im": 0.5}` — valid JSON without Python eval

QCSchema / QCElemental / PennyLane interoperability via the `molssi_qcschema` module

Examples & Coverage

Guides, Demos & Tutorials Supported

`examples/` — 30+ runnable examples built only on qpubench's own schemas and adapters:

Category	Count	Notes
Guides	20	Each maps to a real qpubench mechanism
Demos	6	End-to-end scripts combining several mechanisms
Tutorials	5	Real molecules, via the open `ADAPT-VQE` algorithm
Top-level	3	Gate-based, MBQC, and QForte VQE quick starts

Full breakdown — what's fully supported vs. partial, and why — in `examples/README.md`.

One Guide, End to End

`examples/guides/vqe_calculator.py` — all three layers in one runnable file:

```python
hamiltonian = toy_hamiltonian()                              # Schema layer
config = AdaptVQEConfig(pool_type="SD", optimizer="BFGS", gradient_threshold=1e-5)

calculator = GenericAdaptVQEEngine(                          # Adapter layer
    hamiltonian=hamiltonian, num_qubits=4, num_electrons=2,
    energy_backend=ToyStatevectorAdapter(), config=config,
)
result, vqa, vqa_result = calculator.run()

# Store layer (elsewhere, same as every example): NDJSONStore(...).save(...)
```

Real output: `HF: -4.000000` → `ADAPT-VQE (3 iters): -4.472136` → `exact: -4.472136` (matches to `1e-12`) — the same three-layer shape as every guide/demo above.

Methodology — Honest by Construction (1/2)

No fabricated chemistry

Illustrative qubit Hamiltonians (`toy_hamiltonians.py`) stay explicitly labeled, never presented as physically accurate. Real molecules: `qforte_vqe_benchmark.py` (He/cc-pVDZ) and `hamiltonian_sources/` — real ab initio Hamiltonians replaced toy ones in 3 tutorials, checked against real reference notebooks.

No random numbers standing in for physics

`StubGateAdapter` returns *random* expectation values — useless as an ADAPT-VQE energy oracle. Built `ToyStatevectorAdapter`: a real, small, dense-matrix statevector simulator instead.

Methodology — Honest by Construction (2/2)

No fabricated API calls either

PsiEmbed, libDMET, and SlowQuant aren't on PyPI. Their adapters call each package's *actual documented API* — checked field-by-field against real GitHub source — behind an `ImportError` guard: real code, honestly unexecuted until installed from source.

```
# HF reference:          -4.000000
# ADAPT-VQE (3 iters):   -4.472136
# exact diagonalization: -4.472136   <- matches to 1e-12
```

Summary

Getting Started

```python
from qpubench import (BenchmarkRunner, NDJSONStore, StubGateAdapter,
                      CircuitSpec, ExecutionOptions, SparsePauliObservable)
import pathlib

circuit = CircuitSpec(
    num_qubits=2, serialized=pathlib.Path("bell_state.qasm").read_text(),
    observables=[SparsePauliObservable(num_qubits=2, terms=[...])],
)
runner = BenchmarkRunner(store=NDJSONStore(pathlib.Path("results.ndjson")))
runner.register(StubGateAdapter(seed=42), name="stub")
record = runner.run(circuit, "stub", ExecutionOptions(shots=4096))
print(f"<ZZ> = {record.result.expectation_values[0].value:.4f}")
```

Engineering Quality Gates

Every push to `main` and every PR runs the full gate set in CI (`.github/workflows/ci.yml`, Python 3.11 + 3.12):

Gate	State
`pytest tests/`	340+ tests, no quantum SDK required
`ruff check`	0 errors (deliberate lazy imports allowlisted per-file)
`mypy --strict src/qpubench`	0 errors (untyped optional SDKs silenced per-module)
Doc/code consistency	`tests/test_docs_consistency.py` fails CI if the schema version or module count drifts between code and docs

Tracking document

`CODE_QUALITY_REVIEW.md` — living record of resolved findings, open ToDos, and conventions (single-source schema version, vendor extension points, no computed values in configs).

Roadmap & Links

Repository: `github.com/mqsdk/qpubench`

Schema v3.0.0 — 38 modules · 7 computing models × 5 qubit modalities · zero quantum SDK deps in core

Contributing — writing an adapter takes ~30 lines:

```
cp integrations/template/backend_adapter_template.py  my_adapter.py
# fill the three TODOs: spec, validate(), run()
```

See `INTEGRATION_GUIDE.md` and `AGENTS.md` for the full development workflow.

Contact: `contact@mqs.dk`